

Wecamp Batch 11

Computer Science Foundations

SelfLearning Research Assignment

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Section 1: Programming Fundamentals

Research Topics

- What is programming? Why do computers need code?
- What are a **program**, a **compiler**, and an **interpreter**?
- Core concepts: variables, data types, loops, conditionals, functions
- Differences between **Java** and **JavaScript**
- What is an algorithm and why it matters?

Required Output

- In your own words, explain:
 - What is an algorithm? Provide a real-life example (e.g., making tea).
 - Explain variables, loops, and conditions as if teaching a friend.
- List **3 key concepts** you learned.

1.1 Research Summary

- Computer programming is the process of designing and building executable instructions that a computer can follow to perform specific tasks. It is a foundational skill in the digital age, enabling the creation of software, applications, and systems that power modern technology.
- A computer program is nothing but a set of instructions (smallest unit of execution) that are used to execute particular tasks to get particular results.
- Computers do not communicate like humans so coding acts as a translator. Code converts human input into numerical sequences that computers understand. Once computers receive these messages, they complete assigned tasks such as changing font colors or centering an image.
- A compiler translates an entire program into machine code before execution.
- Interpreters translate and execute code line-by-line during runtime, facilitating easier debugging.

- Java is an object-oriented programming language and has a virtual machine platform that allows you to create compiled programs that run on nearly every platform. Java promised, "Write Once, Run Anywhere".
- JavaScript is a scripting language used for creating interactive and dynamic web pages, it was created in the first place for DOM manipulation. Earlier websites were mostly static, after JS was created dynamic Web sites were made.
- An algorithm is a step-by-step procedure or formula for solving a problem. Algorithms must be precise, efficient, and designed to handle various scenarios.
- References:
 - + <https://www.geeksforgeeks.org/computer-organization-architecture/what-is-a-computer-program/>
 - + <https://www.computerscience.org/resources/what-is-coding-used-for/>
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 - + <https://www.geeksforgeeks.org/javascript/difference-between-java-and-javascript/>

1.2 Required Output

- Algorithm is a set of steps to solve a problem. For examples, cooking rice need following steps:
 - A. Measure the rice and put it into the pot
 - B. Add water into the pot
 - C. Rinse the rice until the water runs clear
 - D. Combine rice and water
 - E. Place the pot into the cooker and turn on
- Variables like labeled boxes in programming. You can name a box whatever you like and store data inside it for later use. In many languages, you must declare and initialize a variable before using it.
For example:
int a = 10;
int b = 8;
- Loops like wheels doing laps. They are structures that repeat a specific block of code multiple times. This continues as long as a certain condition remains true.
- Conditions are the "decision-makers" of a program, much rules in real-life. For example: If it is raining, take an umbrella; else, wear a hat
- 3 key concepts I learned: Program, Algorithm, Data Structure

1.3 Mini exercise:

- The for loop do here to make the repeat of action calculate the total of linear natural numbers.
- Wrapping logic inside a function to make logic becomes generalization and can be reused in anywhere in the code file/component, to avoid the replication in coding.

For example: If user want to calculate the sum from 1 to 100, he/she just called the function sumTo(100).

Section 2: Operating System Fundamentals

Section 2 - Operating Systems (OS) Fundamentals

Research Topics

- What is an **Operating System**?
- Difference between **process** and **thread**
- **Memory management**: RAM vs storage, virtual memory
- **CPU scheduling** (high level)
- What is a **file system**?
- Compare Windows, macOS, and Linux (at a high level)

Required Output

- In your own words, explain:
 - What happens when you double-click to run a program?
 - What does RAM do, and why is it important?
 - Process vs thread — how are they different?

2.1 Research Summary

- Operating System is a software that manages and handles hardware and software resources of a computing device
- Process is a program that is currently in execution within an operating system. It operates in an independent environment and is managed by the OS for proper scheduling and execution. Processes form the basis of program execution in a multitasking system.
- Thread is a smallest unit of execution within a process. It enables a program to perform multiple tasks concurrently while sharing the same memory and resources. Threads improve application performance and responsiveness in multitasking environments.
- Memory management is the process of controlling and organising a computer's memory by allocating portions, called blocks, to different executing programmes to improve the overall system performance.
- CPU scheduling is a process used by the operating system to decide which task or process gets to use the CPU at a particular time. This is important because a CPU can only handle one task at a time, but there are usually many tasks that need to be processed.
- File systems are a crucial part of any operating system, providing a structured way to store, organize and manage data on storage devices such as hard drives, SSDs and USB drives.
- Windows, Linux, and macOS are three prominent operating systems, each with its own strengths and weaknesses.
- Windows is an operating system designed by Microsoft to be used on a standard x86 Intel and AMD processors. It provides an interface, known as a graphical user

interface(GUI) which eliminates the need to memorize commands for the command line by using a mouse to navigate through menus, dialog boxes, buttons, tabs, and icons. The operating system was named windows since the programs are displayed in the shape of a square. This Windows operating system has been designed for both a novice user just using at home as well as for professionals who are into development.

- The Linux OS is an open source operating system project that is a freely distributed, cross-platform operating system developed based on UNIX. This operating system is developed by Linus Torvalds. The name Linux comes from the Linux kernel. It is basically the system software on a computer that allows apps and users to perform some specific task on the computer. The development of Linux operating system pioneered the open source development and became the symbol of software collaboration.
- macOS is a proprietary, Unix-based operating system developed by Apple Inc. It is the primary operating system for Apple's Mac computers and laptops. It was first introduced in 2001 as Mac OS X, and was later renamed to macOS in 2016.
- References:
 - + <https://www.geeksforgeeks.org/operating-systems/operating-systems/>
 - + <https://www.geeksforgeeks.org/operating-systems/difference-between-process-and-thread/>
 - + <https://www.geeksforgeeks.org/operating-systems/memory-management-in-operating-system/>
 - + <https://www.geeksforgeeks.org/operating-systems/cpu-scheduling-in-operating-systems/>
 - + <https://www.geeksforgeeks.org/operating-systems/file-systems-in-operating-system/>
 - + <https://www.educative.io/answers/differences-between-windows-macos-and-linux-operating-systems>
 - + <https://www.geeksforgeeks.org/operating-systems/different-operating-systems/>

2.2 Required Output

- When double-click to run a program, that when you tell the operating system (OS) to execute an application by locating its file on the disk, creating a new process in memory, and loading its machine code into RAM to be run by the CPU.
- RAM stands for Random Access Memory, it is a short-term memory in computer that stores data for active applications and immediate processor use.
- A process is an program in execution with its own dedicated memory space, while a thread is a lightweight unit of execution within a process that shares memory with other threads.
- 3 key concepts I learned: Operating System, Memory, File System

2.3 Mini exercise

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Asus>tasklist

Image Name                      PID Session Name        Session#    Mem Usage
=====
System Idle Process             0 Services              0             8 K
System                          4 Services              0          2,040 K
Secure System                   140 Services            0         56,080 K
Registry                        184 Services            0         24,232 K
smss.exe                        708 Services            0           648 K
csrss.exe                       856 Services            0          2,836 K
wininit.exe                     1100 Services            0          2,012 K
services.exe                   1176 Services            0          9,584 K
lsass.exe                       1196 Services            0          1,700 K
lsass.exe                       1204 Services            0         20,920 K
svchost.exe                     1408 Services            0         34,660 K
fontdrvhost.exe                1440 Services            0           804 K
WUDFHost.exe                   1488 Services            0         14,136 K
svchost.exe                     1576 Services            0         20,652 K
svchost.exe                     1624 Services            0          6,572 K
WUDFHost.exe                   1700 Services            0          3,772 K
svchost.exe                     1804 Services            0          1,668 K
svchost.exe                     1820 Services            0         10,320 K
svchost.exe                     2008 Services            0          7,308 K
svchost.exe                     912 Services            0          6,920 K
svchost.exe                     676 Services            0          5,940 K
svchost.exe                     1096 Services            0         11,288 K
svchost.exe                     1228 Services            0          5,240 K
```

- Running processes are instances of computer programs that is running on my computer. When running tasklist, Window will list all execute file (.exe) that located in RAM and processed by CPU
- When typing tasklist, the screen will display
 - + Image Name : the filename of the executable program currently running
 - + PID: the unique ID of a process
 - + Session Name: this indicates the environment where the process is running
 - + Session #: the specific ID of the user session
 - + Mem Usage: shows the amount of physical RAM the process is currently occupying

Section 3: Networking Fundamentals

Research Topics

- What is the Internet (very high-level overview)?
- Basics of **IP**, **DNS**, and **HTTP**
- Frontend Backend communication at a high level
- What is an **API**? What is a **request** and a **response**?

Required Output

- In your own words, explain:
 - What happens when you type **google.com** into your browser?
 - What is an **HTTP request**?
 - Difference between **IPv4** and **IPv6** (simple comparison)

3.1 Research Summary

- The Internet is a global system that connects computers and devices worldwide, enabling communication, information sharing, and access to digital services.
- IP (Internet Protocol) handles addressing and routing of data packages
- DNS (Domain Name System) translates human-readable domain names into IP addresses (like [google.com](https://www.google.com) to 142.250.190.78)
- HTTP (Hypertext Transfer Protocol) is protocol used to transfer web pages between a server and your browser
- The communication between Frontend and Backend is through special methods. Some common methods can be listed are: RESTful APIs, WebSockets, Server-Side Rendering or GraphQL. However, at a high level, we can know that Frontend and Backend can connect together through Client-Server architecture.
- Client: A device or program that requests data or services (e.g., web browser).
- Server: A system that stores resources, manages data, and responds to client requests.
- Request–response mechanism: Communication follows a structured cycle — client requests, server responds.
- Centralized management: Data and services are controlled from servers, improving security and consistency.
- API stands for Application Programming Interface. They are the invisible backbone of modern software development. They enable applications and systems to communicate and share data efficiently
- APIs operate through a request response cycle between a client and a server
 - + Request: The client sends a request to an API endpoint (URI).
 - + Processing: The API forwards the request to the server.
 - + Response: The server processes and sends back the requested data.
 - + Delivery: The API returns the server's response to the client.
- References:

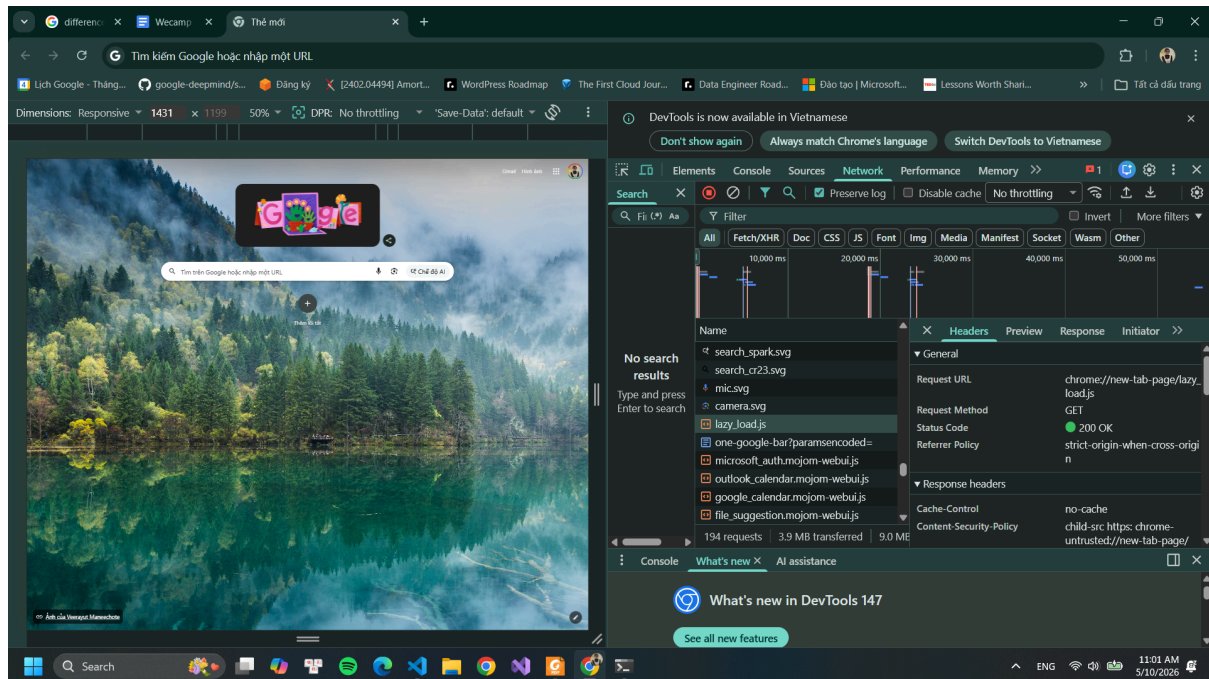
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- + <https://www.geeksforgeeks.org/software-testing/what-is-an-api/>

3.2 Required Output

- Typing [google.com](https://www.google.com) in browser, it display a Google website on my screen
- HTTP Request is a digital message sent by a client to a server to request a resource or action, forming the foundation of web communication
- Difference between IPV4 and IPV6:
- 3 concepts I learn: Network Protocol, IP Address, MAC Address

Feature	IPv4	IPv6
Address Size	32-bit	128-bit
Format	Numeric (e.g., 192.168.1.1)	Alphanumeric (e.g., 2001:db8::1)
Address Count	~4.3 billion	~340 undecillion
Configuration	Manual or via DHCP	Self-configuration (SLAAC) or DHCPv6
Security	Optional (IPsec support)	Built-in (IPsec is integrated)

3.3 Mini Exercise:



- I see method GET when F5 the web
- Status code 200 indicate the request status is success else if 404 is error

Section 4: Database Fundamentals

Research Topics

- What is a **Database** and when do we use it?
- Difference between **SQL** and **NoSQL**
- Tables, records (rows), columns, **primary key**
- **CRUD** (Create, Read, Update, Delete)
- Common DBs: MySQL, PostgreSQL, SQLite, MongoDB

Required Output

- In your own words, explain:
 - Why do applications need databases?
 - SQL vs NoSQL — when might you use each?

4.1 Research Summary

- Data refers to raw, unorganized facts and figures that can be processed to generate meaningful information. A database is a structured system used to store, manage, and retrieve data efficiently for multiple users and applications
- SQL and NoSQL are two different database technologies, with SQL databases emphasizing structured, relational models, and NoSQL databases adopting a flexible,

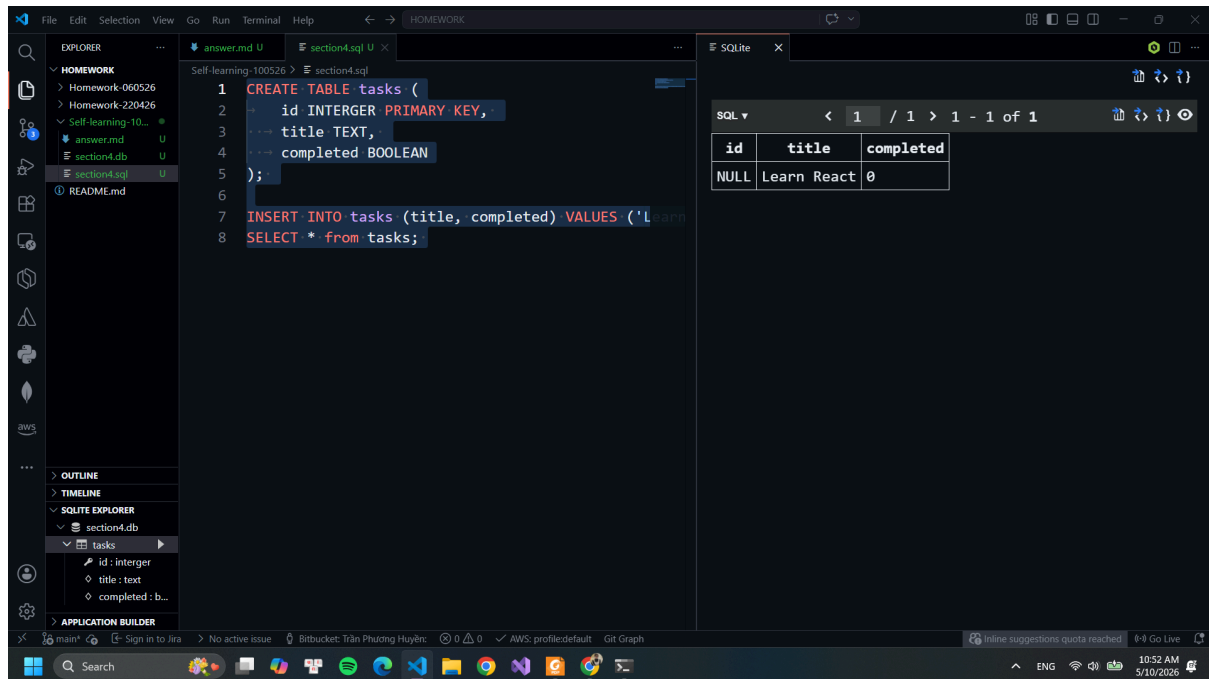
schemaless approach for handling diverse and dynamic data with more emphasis on scalability and speed.

- In the relational database model, each “spreadsheet” is a **table** that stores information, represented as columns (attributes) and **rows (records or tuples)**. **Attributes (columns)** specify a data type, and each record (or row) contains the value of that specific data type. All tables in a relational database have an attribute known as the **primary key**, which is a unique identifier of a row, and each row can be used to create a relationship between different tables using a **foreign key**—a reference to a primary key of another existing table.
- The full form of CRUD is Create, Read, Update, and Delete. These four basic functions represent the essential types of operations that can be performed on data stored within a database.
- The "Create" operation refers to the ability to add new records or data entries into a database.
- The "Read" operation encompasses the retrieval of data from a database.
- The "Update" function pertains to modifying existing data within the database.
- The "Delete" operation refers to the removal of data from a database.
- References:
 - + <https://www.geeksforgeeks.org/dbms/what-is-database/>
 - + https://www.datacamp.com/blog/sql-vs-nosql-databases?utm_cid=19589720824&utm_aid=152984014494&utm_campaign=230119_1-ps-other~dsa-tofu~al_l_2-b2c_3-apac_4-prc_5-na_6-na_7-le_8-pdsh-go_9-nb-e_10-na_11-na&utm_loc=9199204-&utm_mtd=-c&utm_kw=&utm_source=google&utm_medium=paid_search&utm_content=ps-other~apac-en~dsa~tofu~blog~sql&gad_source=1&gad_campaignid=19589720824&gbraid=0AAAAADQ9WsFdDnRuefQ43zKnBe-2wogE_&gclid=CjwKCAjwvvpPBhBuEiwAPMijr8GLFyvr2cmUhmiTDoSMfjJwDhC9O0RqQHfTjOmD1CBkXmx_TjVqcRoChjIQAvD_BwE
 - + <https://cloud.google.com/learn/what-is-a-relational-database>
 - + <https://www.geeksforgeeks.org/websites-apps/crud-full-form/>

4.2 Required Output

- Applications need database for storing, managing and retrieving important data, ensuring that data won't be lost when shut down server. Without a database, data exists only in the computer's RAM. If the application closes or the server restarts, all that information vanishes.
- SQL used for relational database (data is structured, consistent, and requires complex relationships) and NoSQL used for non-relational database (large-scale data, rapid development, and flexible data models).

4.3 Mini Exercise



- First statement: Create a table in database name tasks with properties (id, title, completed)
- Second statement: Statement to insert a row into table tasks with values title = “Learn React”, Completed = False
- Third statement: Statement to select all the rows in the table tasks
- We need primary key to define the unique of each row